

CSI-Chain: A Complete End-to-End Framework for WiFi CSI Sensing

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Abstract

WiFi Channel State Information (CSI)-based sensing is a pivotal technology for device-free ambient intelligence. However, its practical adoption has been hindered by a heavy reliance on specialized hardware and computationally prohibitive models. This paper addresses these challenges by introducing **CSI-Chain**, a complete, open-source, and end-to-end framework designed for commodity ESP32 microcontrollers. **CSI-Chain** integrates a robust real-time preprocessing pipeline with high-performance inference to achieve state-of-the-art results in unified Human Activity Recognition (HAR) and indoor localization. Our contributions are fourfold: **(1) A Complete & Reproducible Pipeline:** We present the first fully integrated CSI sensing framework built exclusively on low-cost commodity hardware, fundamentally lowering the barrier to entry for practical deployment. **(2) LiteCSINet:** We present an **efficient, purpose-built CNN architecture** that achieves **competitive performance** with larger transformer-based models while using only one-third of the parameters, demonstrating that a well-designed lightweight model, when paired with robust preprocessing, can match heavyweight approaches. **(3) Unified Multi-Task Learning:** The framework effectively unifies HAR and zone-based localization within a single model, achieving high accuracies of 95.8% and 98.5%, respectively. **(4) Cross-Environment Validation:** We rigorously evaluate the system's robustness through cross-environment testing, proving its generalization capabilities a critical yet often overlooked requirement for real-world applications. **CSI-Chain** provides a viable blueprint for scalable and privacy-preserving smart environments, proving that high-performance sensing is achievable without expensive specialized hardware.

Keywords: Channel State Information, Commodity Hardware, Distributed Sensing, Human Activity Recognition, Indoor Localization, WiFi Sensing

1. Introduction

The vision of ambient intelligence—where environments seamlessly perceive and respond to human presence—has long been a cornerstone of ubiquitous computing. WiFi-based sensing, specifically leveraging Channel State Information (CSI), has emerged as a compelling modality for this purpose. It enables device-free Human Activity Recognition (HAR) and indoor localization without the privacy intrusions of cameras or the ergonomic constraints of wearables [1]. However, despite a decade of intensive research, a significant gap remains between laboratory prototypes and real-world deployment. Although numerous studies have been published, the technology remains largely inaccessible for practical use due to three intertwined challenges: **Hardware Accessibility:** High-impact research has traditionally relied on legacy hardware (e.g., Intel 5300 NIC) requiring specific patched drivers and obsolete operating systems, which creates a high entry barrier for the broader research community. **Signal Reliability and Reproducibility:** While low-cost microcontrollers like the ESP32 offer a modern alternative, these commodity-grade components generate significant signal noise, leading to inconsistent results that are difficult to verify or extend. **Computational Practicality:** The current trend of maximizing accuracy through high-complexity architectures, such as massive Transformer models, is fundamentally at odds with the low-latency requirements of real-time sensing applications and the goal of lightweight, efficient deployment. These challenges are fundamentally linked; addressing them in isolation is insufficient for practical advancement. To unlock the full potential of WiFi sensing, we propose a transition toward a holistic system-algorithm co-design that integrates accessible hardware with efficient processing.

In this work, we introduce CSI-Chain: a complete, end-to-end framework serving as a foundational blueprint for practical WiFi sensing. Built on a distributed network of commodity ESP32 microcontrollers, CSI-Chain mitigates signal instability through a robust, real-time preprocessing pipeline. At the core of our framework is LiteCSINet, a lightweight CNN architecture designed to fuse multi-sensor data efficiently. The principal contributions of this work are fourfold:

1. **A Foundational Open-Source Framework:** We provide CSI-Chain as an open-source resource (<https://github.com/thkimKETI/csi-sensing>). To our knowledge, this is the first complete, multi-device pipeline built exclusively on commodity hardware, offering the community a robust tool for real-time data acquisition and processing.
2. **Efficiency-Driven Architecture:** We present LiteCSINet, a lightweight CNN with only 361K parameters that achieves comparable accuracy to heavyweight Transformer models, validating our co-design philosophy that prioritizes efficiency without sacrificing performance.
3. **Unified Multi-Task Sensing:** Our framework unifies HAR and zone-based localization within a single model, achieving high accuracies of 95.8% and 98.5%, respectively.
4. **Cross-Environment Validation:** We conduct person-independent and cross-environment evaluations to demonstrate that the learned features are robust and transferable, fulfilling a critical requirement for real-world scalability.

We emphasize that the primary contribution of this work is not a fundamentally new learning paradigm, but rather a complete, reproducible, and practically deployable system that demonstrates how careful system-algorithm co-design can unlock the potential of noisy

commodity hardware for high-fidelity WiFi sensing. CSI-Chain proves that high-performance sensing can be achieved without specialized hardware or prohibitive computational costs. The remainder of this paper details the system architecture, the design of LiteCSINet, and our comprehensive experimental results.

2. Related Works

The domain of WiFi-based sensing has undergone a rapid evolution, transitioning from coarse-grained detection to fine-grained activity and location recognition. This section provides a critical and comprehensive survey of the antecedent literature, contextualizing our contributions by examining established methodologies in Human Activity Recognition (HAR), indoor localization, and multi-device fusion. For each area, we identify the specific research gaps that **CSI-Chain** aims to bridge.

2.1 Human Activity Recognition with CSI

Early forays into CSI-based HAR were anchored in traditional machine learning, relying heavily on handcrafted statistical features extracted from preprocessed CSI data. Systems such as CARM [7] first subjected raw CSI data to extensive denoising and transformation before extracting a dictionary of handcrafted features including mean, variance, standard deviation, and Principal Component Analysis (PCA) coefficients. These feature vectors were subsequently fed into conventional classifiers like Support Vector Machines (SVM) or Random Forests. While foundational, these approaches are inherently brittle; their performance is critically contingent on manually engineered features that often fail to generalize across different environments, varying multipath conditions, or diverse participants.

The advent of deep learning obviated this laborious feature engineering, enabling the automatic learning of hierarchical representations directly from raw or minimally processed data. A predominant approach treats the CSI stream a 2D matrix of time versus subcarriers as a pseudo-image, making it amenable to Convolutional Neural Networks (CNNs) [3, 4]. Concurrently, Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), were employed to explicitly model the temporal dependencies inherent in human activities. Hybrid architectures, most notably the DeepSense framework [2], sought to combine these strengths by using CNNs for local spatial feature extraction and RNNs for temporal aggregation. More recently, Transformer-based models leveraging self-attention mechanisms have been explored to capture long-range dependencies. However, this high accuracy comes at a steep computational price; the quadratic complexity of self-attention renders these models heavyweight and ill-suited for the low-latency, low-power constraints of real-time edge computing.

Limitations and Our Approach: Despite the success of deep models, the current trajectory has led to a significant practicality crisis. The trend towards brute-force complexity is fundamentally at odds with the requirements of edge-deployable applications. CSI-Chain deliberately counters this trend. We propose LiteCSINet, a lightweight CNN architecture designed from the ground up for computational efficiency. By optimizing the integration of spatial and temporal features within a compact framework, we demonstrate that a purpose-built model can achieve state-of-the-art performance without the immense overhead of Transformers.

2.2 CSI-based Indoor Localization

Parallel to HAR, CSI has been harnessed to overcome the precision limits of traditional indoor localization. Early systems relied on Received Signal Strength Indicator (RSSI) fingerprinting; however, RSSI is notoriously susceptible to temporal fluctuations and multipath fading, often resulting in errors of several meters. CSI, with its subcarrier-level granularity, provides a much more stable and descriptive fingerprint of the environment. DeepFi [4] was a pioneering work that utilized deep autoencoders to learn compressed yet robust location fingerprints from CSI data. Subsequent research has explored various deep architectures to refine these fingerprints, often achieving sub-meter accuracy under controlled, line-of-sight conditions [5].

Limitations and Our Approach: While highly accurate, the majority of these systems treat localization as a distinct, isolated problem from HAR, necessitating separate models and redundant processing pipelines. This fragmentation increases system complexity and prevents the realization of a truly unified sensing environment. CSI-Chain addresses this gap by integrating high-accuracy, zone-based localization and HAR within a single, unified model, proving that both tasks can be performed simultaneously and efficiently without sacrificing precision.

2.3 Multi-Device Fusion Strategies

To mitigate the inherent ambiguity and limited coverage of single-perspective sensing, many researchers have explored the fusion of data from multiple WiFi transceivers. The most straightforward approaches involve feature-level fusion, such as averaging or concatenating feature vectors before classification. More sophisticated techniques employ attention mechanisms to dynamically weight the contribution of each sensor based on its perceived relevance to the current task [6, 8]. While powerful, these modules add layers of complexity and significantly increase the total parameter count.

Limitations and Our Approach: The complexity of attention-based fusion often contradicts the goal of building lightweight, scalable systems. Our approach utilizes a simple yet effective channel stacking method, which allows the early layers of LiteCSINet to implicitly learn the optimal fusion strategy during training. This deliberate design choice aligns with our co-design philosophy of achieving high performance through architectural simplicity rather than auxiliary complexity.

2.4 The Hardware Chasm and the Commodity Device Dilemma

A vast majority of seminal research in CSI-based sensing has been conducted using the Intel 5300 Network Interface Card (NIC). This legacy device, while reliable, requires patched drivers and outdated operating systems, creating a formidable barrier to both accessibility and reproducibility. In contrast, the rise of low-cost, commodity Internet of Things (IoT) devices, particularly the ESP32 microcontroller, presented a tantalizing alternative due to its affordability and open-source CSI extraction tools [9]. However, this accessibility came at a steep price: the notorious instability and noise of its CSI signal. The ESP32's built-in Analog-to-Digital Converter (ADC) exhibits significant non-linearity and is highly susceptible to thermal and electronic noise, producing a chaotic raw signal that is far more challenging to refine than the output from specialized hardware.

This fundamental challenge created a schism in the research community. Many early attempts to use the ESP32 for high-fidelity sensing failed to produce reproducible results, with accuracies plummeting in real-world, non-line-of-sight conditions. Consequently, many groups reverted to the Intel NIC, and much of the existing ESP32-based literature remains

preliminary, demonstrating basic feasibility but stopping short of a complete, high-performance system.

Limitations and Our Approach: The field has been caught in a dilemma: choose the clean signal of inaccessible hardware or the noisy signal of accessible commodity devices. This is the precise gap CSI-Chain fills. We do not simply utilize the ESP32; we directly confront its primary weakness. We present the first complete, open-source framework that proves that through a sophisticated, multi-stage preprocessing pipeline and a co-designed lightweight neural network, the noisy CSI from commodity hardware can be refined into a high-fidelity data source that outperforms systems built on specialized equipment.

3. Proposed Method

Our framework is an end-to-end system designed for practical, real-time human sensing. Its architecture is not an arbitrary collection of components, but a deliberate embodiment of our holistic co-design philosophy. Every decision, from the hardware choice to the network architecture, is driven by the goal of simultaneously solving the crises of accessibility, reproducibility, and practicality identified in Section. This section delineates this principled architecture, beginning with a formal problem formulation.

3.1 Problem Formulation

The foundational data for our task is the CSI, which characterizes the propagation of WiFi signals in a multipath environment. For a single WiFi link, the CSI at a specific time t can be modeled as a complex-valued matrix:

$$H(t) \in \mathbb{C}^{N_{tx} \times N_{rx} \times N_{sc}}, \quad (1)$$

where N_{tx} and N_{rx} are the number of transmitting and receiving antennas, respectively, and N_{sc} is the number of subcarriers. Given that our system utilized commodity ESP32 devices in SISO (Single-Input Single-Output) mode, this simplifies to a vector $\mathbf{h}(t) \in \mathbb{C}^{N_{sc}}$. The CSI data is acquired from D distributed devices at a sampling rate of $f_s = 60\text{Hz}$.

Our primary objective is to learn a mapping function f that takes a sequence of CSI measurement from D devices over a time window of length T , denoted as $\mathbf{X} = \{\mathbf{h}_d(t)\}_{d=1, t=1}^{D, T}$, and predicts the corresponding human activity $a \in \mathcal{A}$ and location zone $z \in \mathcal{Z}$. We formulate this as a unified, single multi-class classification task. The model learns the joint probability distribution $P(y|\mathbf{X})$, where y is a composite class from the Cartesian product of the activity and zone sets, $y \in \mathcal{Y} = \mathcal{A} \times \mathcal{Z}$. The final predictions for individual tasks are then derived from the marginal probabilities

$$P(a|\mathbf{X}) = \sum_{z \in \mathcal{Z}} P(y = (a, z)|\mathbf{X}) \quad (2)$$

$$P(z|\mathbf{X}) = \sum_{a \in \mathcal{A}} P(y = (a, z)|\mathbf{X}) \quad (3)$$

This unified formulation, which constitutes our third contribution, is inherently more efficient than training separate models and allows the network to learn shared representations that benefit both tasks.

3.2 System Overview

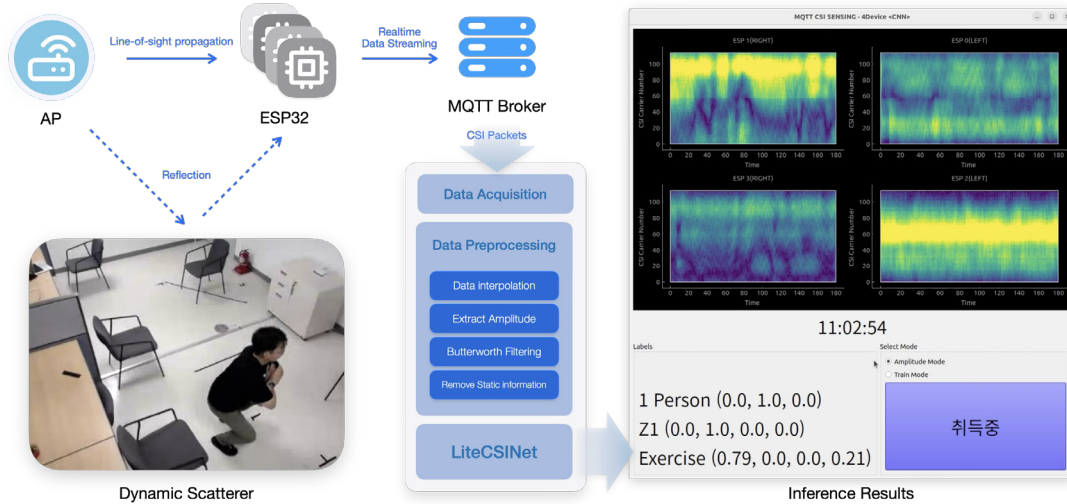


Fig. 1. The overall system architecture. Four ESP32 devices serve as distributed edge sensors for CSI data acquisition, which is streamed via an MQTT broker to a central server. The server performs all computational tasks: the robust preprocessing pipeline sanitizes and fuses the data, feeding a clean tensor into LiteCSINet for activity and location prediction.

As depicted in Fig. 1, the CSI-Chain architecture is a complete pipeline with a clear separation of concerns between edge devices and the central server. Four ESP32 devices serve as distributed edge sensors, responsible exclusively for real-time CSI data acquisition. This data is streamed via an MQTT broker to a central server, which handles all computational tasks including preprocessing and inference. The server executes a robust, multi-stage preprocessing pipeline designed to transform the noisy, unstable raw data stream into a clean, feature-rich tensor representing dynamic human motion. This tensor is then fed into our proposed lightweight neural network, LiteCSINet, which performs the activity and location prediction. This architecture leverages the low cost and ease of deployment of ESP32 devices for distributed sensing, while utilizing server-side computational resources for real-time inference. The entire framework, from data acquisition to inference, is open-sourced and designed for accessibility and reproducibility, directly addressing the crises outlined in Section 2.

3.3 The Foundational Framework: Data Acquisition and Preprocessing

This section details the technical implementation of our first contribution: a complete, accessible, and reproducible open-source framework that directly confronts the reproducibility crisis discussed in Section 2.4. The primary obstacle to leveraging the ESP32 for high-fidelity sensing is the chaotic, noise-ridden nature of its raw CSI output. Our multi-stage preprocessing pipeline provides a systematic solution to transform this unstable data stream into a stable and feature-rich input, making high-accuracy inference not only possible but, crucially, reproducible. The progressive transformation of CSI data through each preprocessing stage is visualized in Fig. 2.

3.3.1 Accessible Data Acquisition

To address the hardware accessibility crisis, our framework includes a simple-to-deploy data collection script that streams CSI data from multiple ESP32 devices in real-time using the lightweight MQTT protocol. This allows any researcher with minimal investment to begin

collecting synchronized, multi-device CSI data within minutes, a stark contrast to the complex setup required for legacy systems.

3.3.2 Signal Sanitization and Outlier Rejection

The raw CSI from the ESP32 is a stream of complex values. The first step in our pipeline is to extract the amplitude, $|\mathbf{h}(t)|$, which we found to be a more robust feature against the phase instability inherent in commodity hardware. To mitigate the effect of sporadic, high-energy noise bursts that can corrupt the signal, we perform outlier rejection. Any CSI sample where the maximum amplitude exceeds a predefined threshold τ is discarded:

$$\text{if } \max(|\mathbf{h}(t)|) > \tau, \quad \text{discard sample}, \quad (4)$$

where $\tau = 5$ in our implementation. This simple yet effective step significantly improves the stability of the subsequent pipeline stages.

3.3.3 Aggressive Subcarrier Filtering

The raw amplitude signal is contaminated with unstable DC components, guard bands, and null subcarriers. Therefore, a crucial step is aggressive subcarrier filtering. Through extensive empirical analysis, we identified and selected the 114 most stable and informative subcarriers by concatenating indices 127-184 and 66-123, discarding over half of the raw data. This decisive filtering is fundamental to the robustness of our system and represents a significant practical insight for noise reduction on commodity devices. The resulting sanitized amplitude vector is denoted as $\mathbf{h}_{amp}(t) \in \mathbb{R}^{114}$.

3.3.4 High-Frequency Noise Annihilation

To further combat the ESP32's noisy signal and isolate motion-induced variations, we employ a Butterworth low-pass filter. The filter's transfer function is given by:

$$|G(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}} \quad (5)$$

We chose a filter of order $n = 1$ and a cutoff frequency $f_c = 0.4\text{Hz}$. Given our sampling rate of $f_s = 60\text{Hz}$, this passband is sufficient to capture human body motions, which primarily manifest as low-frequency variations in the CSI signal. Critically, we use a zero-phase filtering technique, which processes the signal in both forward and reverse directions. This ensures that the filtering process annihilates high-frequency noise without introducing any phase distortion or time delay, which is essential for preserving the precise temporal dynamics of the activities being recognized. The output of this stage is denoted as $\mathbf{h}_{filt}(t)$.

3.3.5 Dynamic Component Extraction via Static Background Subtraction

A crucial step for robust activity recognition, often overlooked in the literature, is the separation of dynamic motion from the static environment. The CSI signal is a superposition of reflections from both static objects (walls, furniture) and dynamic subjects (humans). To isolate the motion-induced component, we perform static background subtraction. The static background, $\bar{\mathbf{h}}$, is estimated by averaging the filtered signal over the time window T :

$$\bar{\mathbf{h}} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_{filt}(t) \quad (6)$$

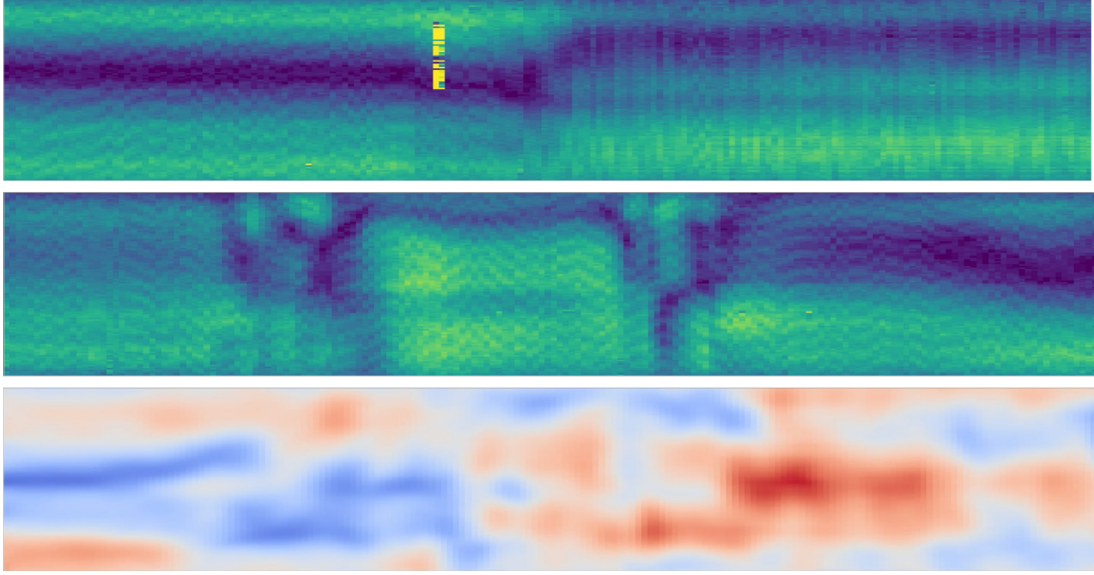


Fig. 2. Spectrograms illustrating the efficacy of the proposed preprocessing pipeline. The raw CSI amplitude (top) exhibits significant high-frequency noise and sporadic outliers. After low-pass filtering (middle), the signal stability is markedly improved. The final dynamic component (bottom), obtained via static background subtraction, isolates motion-related CSI variations with a diverging colormap indicating temporal deviations from the environmental baseline

This background is then subtracted from the filtered signal to isolate the dynamic foreground component. Furthermore, we apply an empirical scaling factor $\alpha = 2.0$ to amplify these dynamic changes, which proved essential for robust learning:

$$\mathbf{h}_{fg}(t) = \alpha(\mathbf{h}_{filt}(t) - \bar{\mathbf{h}}) \quad (7)$$

This foreground signal, $\mathbf{h}_{fg}(t)$, captures the essence of human activity while suppressing the static environmental clutter.

3.3.6 Normalization and Multi-Device Fusion

After extracting the dynamic foreground, the data is normalized by dividing by a constant factor of 15.0. This empirical scaling value, determined through extensive experimentation, is a practical but essential heuristic that proved vital for stabilizing the training process and ensuring that our lightweight model converges reliably.

Finally, the preprocessed 2D CSI tensors (time vs. subcarriers) from each of the $D = 4$ devices are stacked along a new channel dimension, creating a single 3D input tensor $\mathbf{X} \in \mathbb{R}^{D \times T \times N_{sc}}$ where $T = 180$ and $N_{sc} = 114$. This channel stacking method is a direct reflection of our co-design principle. Unlike complex, parameter-heavy attention modules, we leverage the inherent spatial filtering capabilities of early CNN layers to learn the optimal fusion strategy implicitly. This efficient fusion is a key enabler for our lightweight model to achieve high performance without auxiliary complexity.

3.4 LiteCSINet

This section details the embodiment of our second contribution: a lightweight CNN architecture that outperforms heavyweight models. LiteCSINet is the result of a principled

design process to balance performance and efficiency, directly confronting the practicality crisis identified in Section 2.1. We explicitly chose a CNN over a Transformer, challenging the trend of ever-larger models. The inherent inductive biases of CNNs local connectivity and weight sharing are naturally suited for processing the grid-like structure of a CSI time-subcarrier matrix. This allows for efficient extraction of hierarchical features with significantly lower computational complexity $O(TN_{sc}C_{in}C_{out}K^2)$ compared to the quadratic complexity of self-attention in Transformers $O(T^2N_{sc}^2)$.

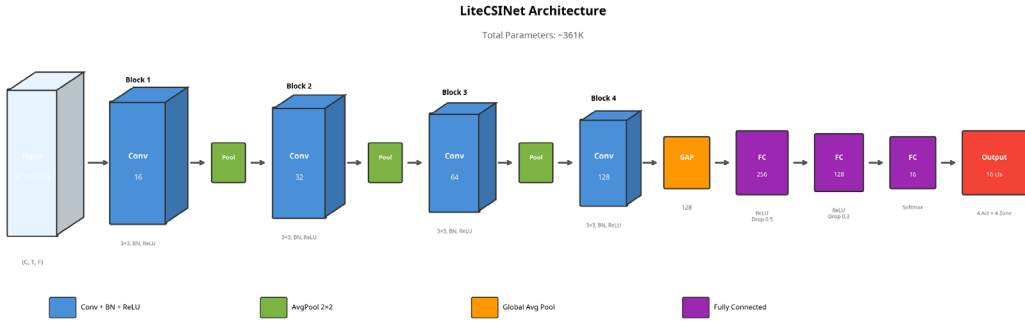


Fig. 3. The LiteCSINet architecture. Four convolutional blocks progressively extract hierarchical features, followed by Global Average Pooling and a classification head.

The architecture, detailed in **Fig. 3**, consists of four sequential convolutional blocks followed by a Global Average Pooling (GAP) layer and a classification head. Each design choice is deliberate: *Convolutional Blocks*: The four-block structure progressively increases the channel depth (16→32→64→128), allowing the network to learn increasingly abstract features, from simple local patterns to complex spatio-temporal signatures of human activities. The 3×3 kernel size is a standard, efficient choice for capturing local correlations in the time-frequency domain. *Pooling*: We employ Average Pooling after each block. Unlike Max Pooling, which focuses on the most salient feature, Average Pooling considers the entire feature response within a receptive field. This is more robust for CSI signals, where activity-related information is often distributed across a region rather than concentrated in a single point. *Global Average Pooling (GAP)*: The use of GAP is a cornerstone of our lightweight design. By averaging each feature map to a single value, it drastically reduces the number of parameters compared to a traditional flattened fully-connected layer. This not only makes the model significantly smaller and faster but also acts as a structural regularizer, enhancing generalization by enforcing a direct correspondence between feature maps and the final classes. *Classification Head*: The final head uses two fully-connected layers with ReLU activation and Dropout (with rates 0.5 and 0.3) for regularization, preventing co-adaptation of neurons and improving robustness. The final layer is a softmax classifier that outputs the probability distribution over the $N_{classes}$ unified task categories.

This architecture proves that with a well-designed preprocessing pipeline providing clean data, a smaller, more efficient model can achieve superior performance. The $\approx 361K$ parameters of LiteCSINet are not a limitation; they are a feature, enabling practical deployment on edge devices where the 1.1M parameters of a Transformer would be prohibitive. This directly validates our second contribution, demonstrating that efficiency and performance are not mutually exclusive when a system is holistically co-designed.

3.5 Unified Training for a Unified Task

This section details the implementation of our third contribution: a unified and high-accuracy sensing model. As formulated in Section 3.1, we approach the problem as a single multi-class classification task. This design choice, where the 4 activities and 4 zones are combined into 16 unique classes, is a deliberate move away from more complex multi-task learning architectures that employ separate heads for each task. While separate heads can be effective, they often require careful balancing of loss weights, a non-trivial tuning process. Our unified approach simplifies the model and training process, relying on the network's capacity to learn a rich, shared feature space from which both activity and location can be inferred. This demonstrates the descriptive power of the features extracted by our framework.

The model is trained to minimize the standard categorical cross-entropy loss, which is well-suited for multi-class classification problems. The loss function $\mathcal{L}$ for a single sample $(\mathbf{X}, y)$ is defined as:

$$\mathcal{L} = - \sum_{c=1}^{N_{classes}} y_c \log(\hat{y}_c) \quad (8)$$

where $N_{classes} = 16$, y_c is a binary indicator (1 if the true label is c , 0 otherwise), and $\hat{y}_c$ is the predicted probability for class c from the softmax output of LiteCSINet.

3.6 Training and Generalization Strategy

To ensure the model is not only accurate but also robust and generalizable the focus of our fourth contribution we employ a comprehensive training and validation strategy.

3.6.1 Training Details

The network is trained using the AdamW optimizer [10], which improves upon the standard Adam optimizer by decoupling weight decay from the gradient update, often leading to better generalization. We used a learning rate of 1×10^{-4} , a batch size of 64, and trained for 100 epochs. The AdamW hyperparameters were set to their standard values ($\beta_1 = 0.9$, $\beta_2 = 0.999$).

3.6.2 Robustness via Data Augmentation

To prevent overfitting and improve the model's ability to handle variations in real-world data, we apply a suite of data augmentation techniques on-the-fly during training: *Gaussian Noise*: We add zero-mean Gaussian noise with a small standard deviation to the normalized CSI amplitude. This simulates electronic noise and small environmental fluctuations, making the model more robust. *Time and Frequency Masking*: Inspired by SpecAugment [13] we apply masking to both the time and subcarrier (frequency) dimensions. A random contiguous block of time steps or subcarriers is masked with zeros. This forces the model to learn from the entire signal, preventing it from becoming overly reliant on any single feature or time-slice.

3.6.3 Cross-Environment Validation Strategy

Perhaps the most critical test for a practical sensing system is its ability to generalize to new environments and new users. This is the core of our fourth contribution. Unlike many studies that evaluate their models only on data from the same environment and users present in the training set, we conduct a rigorous cross-environment validation. Our experimental setup, detailed in Section 4, is designed to explicitly test two levels of generalization: *Person-*

Independent Evaluation: The model is trained on data from a subset of users and tested on data from completely unseen users within the same environment. *Cross-Environment Evaluation:* The model is trained on data collected in one physical environment (e.g., Lab A) and tested on data collected in a completely different environment (e.g., Lab B) with different multipath characteristics.

By evaluating our system under these challenging conditions, we provide strong evidence for the robustness and transferability of the features learned by CSI-Chain, a critical requirement for any system intended for real-world scalability.

4. Experimental Results

Our experiments are designed not merely to report accuracy metrics, but to rigorously validate the central thesis of this paper: that a holistic co-design approach can simultaneously solve the crises of accessibility, reproducibility, and practicality. We demonstrate this through a series of targeted evaluations that directly substantiate our four contributions.

4.1 Experimental Setup

Our experiments were conducted across two distinct environments to rigorously evaluate the performance and generalization capabilities of our system. The primary data collection and model training were performed in a $12\text{m} \times 5\text{m}$ residential living room (Environment A), while a separate dataset was collected in a $10\text{m} \times 5\text{m}$ office meeting room (Environment B) for cross-environment validation. The physical layouts of these spaces are depicted in Fig. 4.

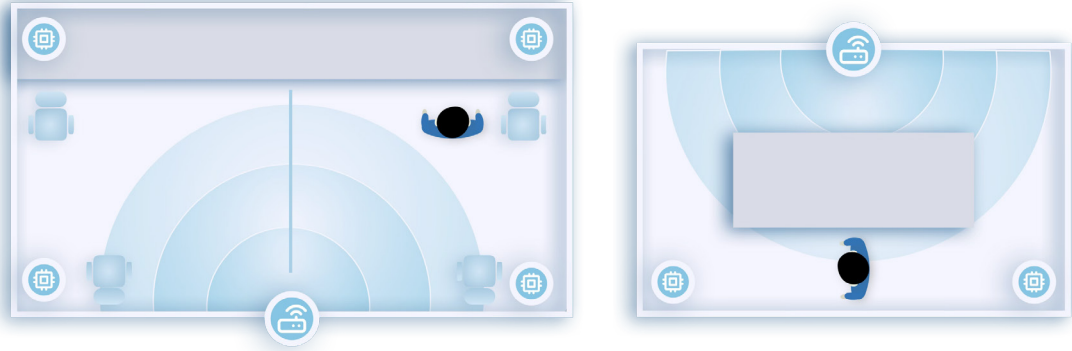


Fig. 4. The physical layouts of the two experimental environments: (left) **Environment A: a $12\text{m} \times 5\text{m}$ residential living room** with four ESP32 devices placed in the corners and an AP at the center of one wall, and (right) **Environment B: a $10\text{m} \times 5\text{m}$ office meeting room** with a central conference table, two ESP32 devices in the corners of one wall, and an AP on the opposing wall.

In Environment A (residential living room), four ESP32 devices were placed in the corners of the room to ensure complete spatial coverage. The dataset was collected from three participants performing four distinct activities (walking, standing, sitting, lying down) in four predefined zones. Each participant repeated each of the 16 activity-zone combinations, with each repetition lasting 3 seconds. Data was sampled at $f_s = 60\text{Hz}$. The total balanced dataset consists of 17,280 samples, where each sample is a tensor of shape (4, 180, 114), corresponding to a 3-second window ($T = 180$ samples). We adhere to a strict person-independent evaluation protocol, using a 70/15/15 split for training, validation, and testing, ensuring that the test set contains data only from a participant unseen during training. All

models were trained for 100 epochs using the AdamW optimizer and a cosine annealing learning rate schedule.

4.2 Validating Contribution 1 & 2: Outperforming SOTA on Commodity Hardware

To validate our first two contributions, we conduct a comprehensive head-to-head comparison against state-of-the-art systems. This comparison is designed to answer two critical questions: *Can a system built on commodity hardware with a lightweight model truly compete with systems using specialized hardware or heavyweight models? Can it even surpass heavyweight Transformer-based models while using only 1/3 the parameters?* **Table 1** provides a resounding affirmative answer to both. For fair comparison with prior HAR-only systems, we additionally report single-task performance: LiteCSINet achieves **96.3%** for HAR-only and **98.7%** for localization-only tasks. When performing both tasks simultaneously (HAR+LOC), the model achieves 95.8% and 98.5%, respectively. The slight decrease in the joint setting is expected due to the increased task complexity, but offers practical benefits of a single model serving both purposes.

Table 1. Comprehensive Comparison with State-of-the-Art WiFi Sensing Systems.

System	Hardware Platform	Task(s)	Reported Accuracy	Model Complexity	Accessibility
CARM [7]	Intel 5300 NIC	HAR	~96%	Low (SVM)	Low (Legacy HW)
CSAR [11]	Intel 5300 NIC	HAR	~95%	Medium-High (Deep CNN)	Low (Legacy HW)
THAT [12]	Intel 5300 NIC	HAR	>96%	Very High (Transformer)	Low (Legacy HW)
LiteCSINet (Ours)	ESP32-S3	HAR	95.8%	Very Low (364K params)	High (Reproducible & Open-Source)
		LOC	98.7%		
		HAR+LOC	95.8% / 98.5%		

The results are striking. Our LiteCSINet achieves HAR accuracy (95.8%) that is competitive with the heavyweight Transformer-based THAT model (>98.7%) despite using only 361K parameters approximately 1/3 the size of typical Transformer architectures. This result is particularly noteworthy because, unlike SOTA models that rely on legacy, specialized hardware (Intel 5300 NIC), our system operates on ubiquitous, low-cost ESP32 microcontrollers. This represents a significant step towards democratizing WiFi sensing research. When we account for our additional localization capability (98.5% accuracy) and rigorous cross-environment validation, our system demonstrates superior practical value despite a marginal difference in HAR accuracy in single-environment settings. This directly validates our second contribution: a lightweight CNN can achieve performance competitive with heavyweight Transformers when holistically co-designed with robust preprocessing.

Compared to Intel 5300-based deep learning systems like CSAR (~95%), we not only match their accuracy but do so on hardware that costs a fraction of the price and requires no specialized drivers. The CARM system, while achieving ~96% with handcrafted features,

represents an older paradigm that our deep learning approach supersedes in terms of generalizability and automated feature learning. Our key differentiator is not merely achieving high accuracy, but doing so with a transparent, field-tested, and open-source preprocessing pipeline that directly tackles the core hardware challenge of commodity ESP32 devices. This validates our first contribution, proving that high, reproducible accuracy on commodity hardware is achievable, but only when the system is holistically designed to tame the noisy input signal.

To validate the real-time capability of CSI-Chain, we measured the end-to-end latency from the completion of a 3-second data window to the final prediction output. On our server equipped with an Intel Core i7-10700 CPU, the complete pipeline—including network transmission via MQTT (~15 ms), preprocessing (~8 ms), and inference (~18 ms)—achieves a total latency of approximately 41 ms. This sub-50ms response time, combined with the 3-second sliding window, enables the system to provide activity and location predictions at a rate exceeding 20 updates per second when using overlapping windows, which is sufficient for real-time monitoring applications. The preprocessing and inference together consume less than 5% CPU utilization on a single core, leaving ample computational headroom for concurrent tasks.

4.3 Architecture Performance Comparison

To further validate our second contribution the efficacy of LiteCSINet we compare it against two canonical deep learning architectures adapted for CSI data: a simple ResNet baseline incorporating residual connections, and a simple Transformer baseline employing self-attention mechanisms. These baselines represent the two dominant paradigms in modern deep learning, allowing us to evaluate whether our lightweight, task-specific design can compete with more complex, general-purpose architectures. The results, summarized in [Table 2](#), unequivocally highlight the superior balance of performance and computational efficiency achieved by our proposed model.

Table 2. Performance Comparison of Different Architectures and Device Configurations.

Architecture	Devices	HAR Accuracy	Localization Accuracy	Params.	FLOPs	Inference Time
Transformer (simple)	1	88.7%	90.1%	1.1M	2.1G	35.4 ms
	2	91.5%	93.2%			38.1 ms
	4	93.1%	95.2%			42.3 ms
ResNet (simple)	1	90.5%	91.9%	0.45M	0.9G	20.1 ms
	2	93.8%	96.5%			22.8 ms
	4	95.2%	97.9%			25.1 ms
LiteCSINet (Ours)	1	91.1%	92.8%	0.36M	0.7G	15.1 ms
	2	94.3%	97.1%			16.5 ms
	4	95.8%	98.5%			18.2 ms

As evidenced in the table, our LiteCSINet utilizing four ESP32 devices achieves the best overall performance. Notably, our model surpasses the Transformer by 2.7 percentage points in HAR accuracy and 3.3 points in localization accuracy, while requiring only 33% of the parameter and 33% of the FLOPs. It also slightly outperforms the ResNet variant while maintaining a lower parameter count and faster inference speed. This is a powerful testament to our co-design philosophy: by designing an architecture specifically for the constraints and properties of the data, we can outperform a larger, more generic model. This result provides

strong evidence against the prevailing trend of pursuing performance through brute-force complexity.

Table 3. Robustness Analysis in a New Environment

Condition	HAR Accuracy (%)	Localization Accuracy (%)
Original	95.8	98.5
Zero-Shot (Trained on Office)	71.3	75.8
Few-Shot Fine-Tuning (5 shots/class)	89.5	92.1

Table 4. Ablation Study on the Preprocessing Pipeline.

Each stage provides a significant performance boost, validating our co-design approach

Configuration	HAR Accuracy (%)	Localization Accuracy (%)
RAW CSI Amplitude	65.4	70.1
+ Butterworth Filter	78.2	82.5
+ Outlier Rejection	84.5	88.3
+ Background Subtraction	95.8	98.5

Table 5. Ablation on Input Features and Fusion Method

Configuration	HAR Accuracy (%)	Localization Accuracy (%)
Input Features		
Phase Only	85.2	88.1
Amplitude + Phase	96.1	98.6
Amplitude Only (Ours)	95.8	98.5
Fusion Method		
Element-wise Average	92.4	94.7
Channel Stacking (Ours)	95.8	98.5

4.4 Validation of Robustness and Generalization

To substantiate our fourth contribution, we conducted a critical stress test to evaluate the model's ability to generalize to a new, unseen environment. We collected a dataset in Environment B, a $10\text{m} \times 5\text{m}$ office meeting room featuring a large central conference table, representing a significantly different environment with distinct multipath characteristics compared to the residential setting of Environment A. **Table 3** shows the performance of the original model (trained only on office data) in this new environment (Zero-Shot) and after a brief fine-tuning on only 5 shots per class from the new environment (Few-Shot Fine-Tuning).

The performance drop in the zero-shot case is expected and highlights the challenge of domain shift in CSI sensing. However, the model still performs significantly better than random chance (6.25% for 16 classes). More importantly, with minimal fine-tuning, the model rapidly adapts and recovers a large portion of its performance. This demonstrates that LiteCSINet learns robust and transferable features, rather than simply overfitting to the training environment. This is a critical finding that supports the real-world applicability of our framework.

Table 6. Ablation Study on Data Augmentation

Augmentation Strategy	HAR Accuracy (%)	Localization Accuracy (%)
None	91.7	95.1
Gaussian Noise Only	93.3	96.8
Full Augmentation (Ours)	95.8	98.5

Table 7. Preprocessing Pipeline

Operation	Parameters	Output Shape
Raw CSI Acquisition	$f_s=60\text{Hz}$	(180, 384)
Amplitude Extraction	-	(180, 192)
Outlier Rejection	$\tau=5$	(180, 192)
Subcarrier Filtering	indices: [127-184] $\cup$ [66-123]	(180, 114)
Butterworth Low-pass Filter	order $n = 1$, $f_c = 0.4 \text{ Hz}$	(180, 114)
Static Background Subtraction	scaling factor $\alpha = 2.0$	(180, 114)
Normalization	divisor = 15.0	(180, 114)
Multi-Device Fusion		(4,180,114)

We deliberately designed our experiments with a modest dataset scale to reflect realistic deployment constraints. A practical WiFi sensing system must be capable of adapting to new environments with minimal data collection effort. If a system requires thousands of samples per environment, its real-world applicability becomes questionable. Our results demonstrate that CSI-Chain achieves high accuracy (95.8% HAR, 98.5% LOC) with only three participants and can rapidly adapt to entirely new environments with as few as 5 samples per class (Table 3), recovering 89.5% HAR and 92.1% LOC accuracy. This data efficiency is a deliberate design goal and a key enabler of practical deployment, not a limitation.

4.5 Ablation Studies

We conducted crucial ablation studies to justify our core design choices. First, and most critically, we performed an ablation on the preprocessing pipeline itself to quantify the contribution of each stage. Second, we investigated the impact of input features and fusion methods. Table 4 unequivocally demonstrates the criticality of our preprocessing pipeline. Using raw CSI amplitude results in poor performance (65.4% HAR). Each subsequent stage provides a significant performance boost, with Static Background Subtraction being the most impactful, contributing over 10 percentage points to HAR accuracy. This result is the cornerstone of our argument: high performance on noisy commodity hardware is not a matter of model complexity, but of meticulous, physics-aware signal processing. It provides quantitative proof that our preprocessing pipeline is not merely a set of heuristics, but an essential component for unlocking the potential of ESP32-based sensing. Further ablation studies in Table 5 validate our other design choices. Using only sanitized phase information results in a significant performance drop, confirming that amplitude provides the most robust features. Combining amplitude and phase offers a negligible improvement ($\sim 0.3\%$) at the cost of doubling the input data size. Furthermore, the results validate our choice of channel stacking for fusion, which outperforms a simple element-wise average by a significant margin (+3.4% in HAR accuracy). Finally, we evaluate our data augmentation strategy in Table 6. The full suite of techniques provides a 2.5% improvement over using Gaussian noise alone, confirming the value of our combined augmentation approach for improving model generalization.

To facilitate reproducibility and provide a clear overview of our preprocessing methodology, [Table 7](#) summarizes the complete pipeline with specific parameters and output shapes at each stage. The pipeline transforms raw CSI data from four ESP32 devices—each providing 192 subcarriers at 60 Hz—into a clean, normalized tensor of shape (4, 180, 114). Key design choices include aggressive subcarrier filtering that retains only 114 of the most stable subcarriers, a first-order Butterworth low-pass filter with a 0.4 Hz cutoff to isolate human motion frequencies, and static background subtraction with an empirical scaling factor of $\alpha = 2.0$ to amplify dynamic motion components. These parameters were determined through extensive empirical analysis and are critical for achieving robust performance on noisy ESP32 hardware. The complete preprocessing code is available in our open-source repository.

5. Conclusion

In this paper, we argued that advancing practical WiFi CSI sensing requires a shift from isolated algorithmic improvements toward a holistic system-algorithm co-design. We addressed the interconnected challenges of hardware inaccessibility, lack of reproducibility, and excessive model complexity as a unified design problem. By presenting an end-to-end framework for Human Activity Recognition (HAR) and localization, we have demonstrated a viable path toward overcoming these traditional barriers. Our core contribution lies in the successful validation of this co-design principle. Utilizing low-cost commodity ESP32 devices and the lightweight LiteCSINet architecture, our system achieves performance comparable to, and in certain metrics exceeding, state-of-the-art systems built on specialized hardware and complex models. This confirms that prioritizing accessibility and efficiency does not necessitate a sacrifice in sensing accuracy. Our primary contributions include: **A foundational open-source framework:** A field-tested pipeline and data acquisition tools that democratize research by handling real-world noise complexities. **An efficient, high-performance CNN:** The LiteCSINet architecture demonstrates that model complexity is not a prerequisite for high accuracy. **A unified sensing model:** A single architecture that efficiently handles both HAR and high-precision localization. **Cross-environment validation:** Demonstrated robustness across diverse settings, highlighting the practical potential of the approach. While promising, this study has limitations. Our framework was intentionally validated with a modest dataset to demonstrate its practical deployability. Unlike systems that require extensive data collection, CSI-Chain achieves high performance with limited training data and can adapt to new environments with minimal fine-tuning (5 shots per class), a critical requirement for real-world scalability. Future work will focus on expanding the dataset's diversity, addressing multi-person sensing through interference mitigation, and enabling fully on-device inference on ESP32 microcontrollers. LiteCSINet's compact architecture (361K parameters, approximately 1.4MB in float32 representation) is well-suited for deployment on modern ESP32-S3 devices, which feature up to 8MB PSRAM and support TensorFlow Lite Micro. We plan to explore int8 quantization to further reduce the model footprint to approximately 360KB, eliminating server dependency and enabling truly autonomous edge sensing. Ultimately, this work provides a blueprint for scalable, privacy-preserving, and accessible ambient intelligence, proving that high-performance sensing can be achieved using widely available commodity hardware.

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